Tushar Kumar Tiwari

+91 8122061903 | tushartiwari91410@gmail.com | linkedin.com/in/tushar-kumar-tiwari | github.com/TusharCodeup

Summary

B.Tech CS student specializing in Agentic AI, RAG architectures, and full-stack ML. Skilled in Python, LangGraph, LangChain, Neo4j, Pinecone, and FastAPI, with hands-on experience building multi-agent pipelines (40% token reduction), dual-retrieval GraphRAG engines (5,500+ node graphs), and real-time ML platforms (78% accuracy).

Technical Skills

Languages: Python, C++, SQL
AI & LLM: LangGraph, LangChain, GraphRAG, Autonomous Agents, Gemini API, Prompt Engineering
ML & Data: Scikit-learn, Pandas, NumPy, Feature Engineering, Predictive Modeling, Vector Embeddings
Backend & Web: FastAPI, REST APIs, Node.js, React.js, HTML5, CSS3
Databases: Neo4j (Graph DB), Pinecone (Vector DB), MySQL
Tools: Docker, Git, GitHub, Google Colab

Projects

Decode Team | *LangGraph, Gemini API, Docker, Node.js, AST* | **GitHub** Aug 2026 – Present

- Architected a LangGraph + Gemini multi-agent system with a cyclic state machine across 5 specialized agents (PM, Architect, Coder, Reviewer, Debugger) to synthesize full-stack apps from raw requirements.
- Engineered a 3-tier diagnostic triage & self-healing loop that eliminated runaway execution, cutting token usage by 40% via <1ms AST syntax gates and convergence circuit breakers.
- Built an ephemeral Docker sandbox enforcing zero-trust security, package whitelisting, and Git-backed rollbacks on test regression.
- Implemented a SHA-256 Content-Addressable Virtual File System to eliminate LangGraph state bloat, passing 100% of a 76-test end-to-end code generation suite.

CineGraph-RAG | *LangChain.js, Neo4j, Pinecone* | **GitHub** Jan 2026 – July 2026

- Built a dual-retrieval GraphRAG system indexing 1,000+ TMDB movies across a Neo4j knowledge graph (5,500+ nodes, 13,600+ relationships) and a 3,072-dim Pinecone vector store.
- Developed an intent-driven LLM router that bifurcates queries into parameterized Cypher graph traversals or Pinecone semantic similarity search.
- Eliminated multi-hop entity hallucinations by grounding Gemini responses with topological graph paths and contextual vector metadata.
- Designed a stateful ingestion pipeline with automated checkpoint recovery and exponential backoff, ensuring API rate limits.

IPL Analytics Platform | *Python, FastAPI, React.js, Scikit-learn, Recharts* | **Live Demo** June 2025 – Dec 2025

- Deployed FastAPI REST endpoints serving real-time match analytics and live win-probability predictions to an interactive React.js dashboard with Recharts visualizations.
- Built Scikit-learn models trained on Cricsheet IPL data, achieving 78% accuracy on live match-situation inference.
- Performed end-to-end preprocessing, feature engineering (run rate, wicket pressure, powerplay metrics), and team-level statistical aggregation to optimize prediction reliability.

Leadership & Initiatives

Competed as a school delegate in the district chess championship at IIT Madras.

Core Member in workshop and event management teams; coordinated cross-functional academic initiatives

Exploring UI/UX trends and creative web designs; enjoy football, badminton, and tech podcasts

Education

Vellore Institute of Technology Bhopal, MP
B.Tech in Computer Science; CGPA: 8.54 *Aug 2023 – May 2027*
Kendriya Vidyalaya No.1 Arakkonam Tamil Nadu
Senior Secondary (12th); Percentage: 78.2% *Aug 2022 – May 2023*
Kendriya Vidyalaya, Ghazipur Uttar Pradesh
High School (10th); Percentage: 94.8% *Aug 2021 – May 2022*

Certifications

Cloud Computing – NPTEL | Internet of Things – NPTEL | Machine Learning Specialization – Stanford